

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-005 - Nahorkatiya-005

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000050
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2017-03-14 / 2017-06-15
Measured / TVD	3165 m / 3126 m
Trajectory	Directional; inc 31.9 deg; az 45.0 deg
Formation Interval	1655-2357 m
Lithology	Fine sandstone with carbonaceous shale
Pore Pressure Reference	3044 psi
Fracture Gradient	15.29 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Critical (67.7/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	791	Surface
9 5/8 in casing	1835	Intermediate
7 in liner/casing	2785	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2017-04-02 812 m	Cementing Issue	Critical	Cement returns were below planned percentage. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Offset loss history should inform cementing hydraulics.
2017-04-25 1553 m	Torque Spike	Critical	High torque developed while WOB remained elevated. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Use offset torque trend to set adaptive operating envelope.
2017-04-14 1215 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Use offset torque trend to set adaptive operating envelope.
2017-05-02 1798 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Avoid prolonged static periods in high-drag interval.
2017-05-06 1919 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2017-05-15 2210 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2017-04-03 849 m	NPT	Medium	Rotary and WOB remained at zero during downtime. Mitigation: Classify NPT cause consistently for offset-well learning. Lesson: Consistent NPT coding enables meaningful cross-well comparison.

Completion Summary

The synthetic well reached 3165 m MD in the Barail interval. The NWIS knowledge record links 15 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.